

Additional Text Changes

Writing Tasks Not Yet Captured in Any Mentor Report

Neural Network Emulation of the Unrestricted Three-Body Problem

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1. Overview and Purpose

Six mentor reports have been generated covering all experimental work. A cross-check against the full list of planned changes revealed six writing tasks that belong naturally to existing mentor reports but were not included in them. These are not new experiments — they are purely textual additions: sentences, paragraphs, or claim statements that the paper needs but which were not written up in the relevant report.

This document adds those missing pieces. For each item, it identifies which mentor report it belongs to, explains the motivation and value to the paper, provides the exact text to add to V3, and explains how to integrate it with the changes already proposed in that mentor report.

#	Missing Item	Belongs to	Paper Section Affected
1	Strong central claim + generalizable principle	Revised Phase 1 report	Section 1 / Introduction
2	Section 3.1 research question update	Revised Phase 1 report	Section 3.1
3	One strong sentence for Introduction	Revised Phase 1 report	Section 1
4	CPU vs GPU explanation (G1)	Step 6 report	Section 4 / Conclusion 5
5	Conclusion 2 full expansion with 105×	Step 6 report	Section 5, Conclusion 2
6	NN invocation rate moved into Results (G3)	Review report	Section 4.1

PART A — Additions to: Revised Phase 1 Mentor Report

The Revised Phase 1 report covers the Vector NN baseline experiment and its implications. It concludes with Section 7 (Integration into V3 Paper), which provides paper modification guidance. Three writing tasks belong here: the central claim statement for Section 1, the updated research question for Section 3.1, and a strong introductory sentence. These were identified as needed but the exact text was not written.

2. Addition A1: Strong Central Claim + Generalizable Principle

2.1 What Is Missing

The Revised Phase 1 report correctly identifies the updated central claim in Section 7.1: *"Why does SIMON maintain stability where Vector NN fails?"* and provides the framing logic in Sections 7.3 and 7.3a. However, it does not provide a single, standalone, quotable sentence expressing the generalizable scientific principle — the kind of sentence that appears at the end of an introduction or at the start of a conclusions section and captures the paper's contribution in one breath.

2.2 Why It Matters for the Paper

Top-tier papers are remembered for one clear claim. The ChatGPT review specifically flagged "one strong claim" as missing from V2. Without it, a reader finishes the paper knowing what SIMON does but not why it matters or what principle it demonstrates. The generalizable principle — that physics-informed architectural constraints are necessary, not merely beneficial, in chaotic neural network force models — is the paper's contribution to the field beyond the specific three-body result.

2.3 How It Fits with the Revised Phase 1 Report

The Revised Phase 1 report's Section 7.1 provides the central claim in question form. Section 7.2 lists the new paper sections needed. The standalone principle sentence below should be added as a boxed addition immediately after Section 7.1, labelled as "Suggested sentence for Section 5 Conclusions and Section 1 Introduction." It directly follows from Finding 2 (ejection vs bounded) and Finding 3 (gate sensitivity analysis) which are already in the report.

2.4 Exact Text to Add to V3

Suggested text for Section 5 (Conclusions) and Section 1 (Introduction)

% Add as the opening sentence of Section 5 (Conclusions) in V3:

We demonstrate that enforcing physics-based constraints in neural force models is necessary, not merely beneficial, for long-horizon orbital stability in chaotic gravitational systems: a neural network trained to sub-degree directional accuracy causes body ejection through accumulated non-physical impulses, while a scalar correction architecture that enforces

analytic force direction by construction maintains bounded orbits for the full 100-year simulation horizon.

% Also add as a condensed version in Section 1 (Introduction), last paragraph:

We show experimentally that this constraint is necessary: removing analytic direction enforcement while retaining comparable force accuracy causes body ejection, while the constrained architecture maintains orbital stability.

3. Addition A2: Section 3.1 Research Question Update

3.1 What Is Missing

The V3 Action Plan (Step 9) explicitly specified updating Section 3.1's research question to reference the direction vs magnitude comparison. The Revised Phase 1 report lists this as a requirement in Section 7.2 ("Abstract (new): Open with the scientific question") but does not provide the exact revised text for Section 3.1 itself.

The current V2 Section 3.1 research question reads: *"How closely can a physically constrained neural network force model approximate the long-term stability of a traditional numerical integrator?"* This is a reasonable question but does not reflect the new experimental contribution.

3.2 Why It Matters

Section 3.1 is where the reader learns what the paper is trying to answer. If the research question is not updated, the Vector NN experiment appears as a disconnected addition rather than the central experimental contribution. The updated question should make the direction vs magnitude comparison the primary question, with ias15 approximation as the secondary question.

3.3 How It Fits with the Revised Phase 1 Report

The Revised Phase 1 report's Section 2.3 already states the central research question in the experimental context: *"Does constraining force direction to analytic geometry provide long-term orbital stability benefits...?"* The paper's Section 3.1 should echo this, not the older V2 formulation. The integration is straightforward: add the updated question as "Revised Section 3.1 text" immediately after Section 7.2 in the mentor report.

3.4 Exact Text to Add to V3

Revised Section 3.1 Research Question for V3

% Replace the research question in Section 3.1 with:

This paper addresses two related questions. The primary question is:

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does structurally enforcing analytic force direction in a neural force
model provide long-horizon orbital stability advantages that an equally
accurate learned-direction model cannot replicate? The secondary question
is: how closely can such a physically constrained model approximate the
trajectory stability of a reference high-order integrator (ias15) over
100-year chaotic three-body simulations, and at what computational cost?

% This replaces the single research question with two, making the
% Vector NN comparison the primary contribution.
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4. Addition A3: Strong Introduction Sentence

4.1 What Is Missing

Related to A1 but distinct: V2's Introduction (Section 1) currently motivates the three-body problem and neural network approaches but does not contain a sentence stating what the paper will prove. A strong final sentence in the introduction that states the result — before the reader has to find it in the conclusions — is standard practice in top-tier venues.

4.2 Why It Matters

The ChatGPT review flagged the need for a “generalizable principle framing.” This is most effectively delivered in the introduction, where the reader expects to learn not just what the paper does but what it proves. Section 1 currently ends by describing the evaluation methodology. It should end by stating the finding.

4.3 How It Fits with the Revised Phase 1 Report

This addition is the introduction-side complement to Addition A1 (the conclusions-side statement). Together, A1 and A3 create a matching framing: the introduction states what will be shown, the conclusions confirm it was shown. Both draw directly from Phase 1's Finding 2 and Finding 3 which are already in the report.

4.4 Exact Text to Add to V3

Add as final paragraph of Section 1 (Introduction) in V3

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% Add as the final paragraph of Section 1 (Introduction) in V3,
% immediately before Section 2:
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In this work, we evaluate whether the physics constraint is doing
real work or merely describing an architectural choice. We introduce
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a controlled Vector NN baseline that learns force direction rather than fixing it analytically, with exact analytic magnitude – giving it strictly more information than SIMON. We show that despite training to 0.04° directional accuracy, the Vector NN causes body ejection through accumulated non-physical perpendicular impulses, while SIMON maintains bounded orbits for the full 100-year horizon. This demonstrates that the physics constraint is necessary: sub-degree directional learning error is sufficient to destabilise long-horizon chaotic gravitational simulations.

PART B — Additions to: Step 6 Mentor Report

The Step 6 report covers adaptive sub-stepping before/after analysis. Its Section 6.2 provides three additions for V3 (new Section 4.4, expanded Conclusion 2, updated Section 3.7 motivation sentence). Two writing tasks were not included: the CPU vs GPU explanation and the complete Conclusion 2 replacement text with all numbers.

5. Addition B1: CPU vs GPU Explanation (G1 from SIMON Report)

5.1 What Is Missing

The SIMON Optimization Report (Gap G1) identified that the paper needs to explain why NumPy on CPU outperforms PyTorch on GPU for $N=3$ bodies. This explanation exists in the SIMON report's technical section but was never carried through into any mentor report as a specific paper addition.

V2 currently attributes the $54\times$ speedup to “changes in processing power (switch from Apple M1 to Nvidia GeForce), changes in network architecture and hyperparameters, and skipping the NN for far field encounters.” This is accurate but does not explain the counterintuitive choice to use CPU NumPy over GPU PyTorch for the inference loop — which a computational reviewer will immediately question.

5.2 Why It Matters

Any reviewer with ML background will ask: “*Why not use the GPU for inference?*” Without an explicit explanation, the paper appears to have made a suboptimal implementation choice. The answer is technically rigorous and actually strengthens the paper: for $N=3$, each CUDA kernel launch costs $\sim 15\mu\text{s}$ of overhead, and the NN forward pass requires ~ 30 small operations — giving $450\mu\text{s}$ of pure overhead per step, compared to $24\mu\text{s}$ for NumPy on CPU. The GPU would only be faster for $N > \sim 50$ bodies where $O(N^2)$ pair computation amortises the kernel launch cost.

5.3 How It Fits with the Step 6 Report

The Step 6 report's Section 6.2 already proposes a revised Conclusion 1 and new Section 4.4. The CPU/GPU explanation belongs in Conclusion 5 of V2, which currently reads: “*Optimising implementation is more important than algorithmic novelty.*” This conclusion is the right home for the CPU/GPU explanation — it transforms a vague claim about implementation into a concrete, technically grounded statement. Add this as “Addition 4 — Expanded Conclusion 5” in the Step 6 report.

5.4 Exact Text to Add to V3

Replace Conclusion 5 in Section 5 of V3

% Replace Conclusion 5 in Section 5 of V3:

`\textbf{Implementation efficiency determines practical performance:}`

The $54\times$ reduction in per-step inference time (from $1\{,\}300\sim\mu\text{s}$ to $24\sim\mu\text{s}$) arose from implementation optimisation rather than

algorithmic changes: switching from PyTorch GPU inference to pure NumPy CPU inference, and reducing hidden layer width from 128 to 32 units. For $N=3$ bodies, each CUDA kernel launch incurs approximately $15\sim\mu\text{s}$ of overhead; with ~ 30 operations per forward pass, GPU inference costs $\sim 450\sim\mu\text{s}/\text{step} \sim 19\times$ slower than NumPy. GPU inference becomes advantageous only for $N \gtrsim 50$ bodies, where $O(N^2)$ pair evaluations amortise the kernel launch cost. This implementation insight generalises to other small- N gravitational NN systems.

6. Addition B2: Conclusion 2 Full Expansion with All Numbers

6.1 What Is Missing

The Step 6 report's Section 6.2 (Addition 2) provides a suggested replacement for Conclusion 2 in a narrative callout box. However, reading it carefully, the text provided is approximately 80 words and covers the key numbers. What is missing is the complete connection to the no-NN ablation result (from the Review report) which established that adaptive stepping, not the NN, is the primary stability mechanism. The Conclusion 2 text should now incorporate this finding for a complete picture.

6.2 Why It Matters

Conclusion 2 in V2 currently states adaptive stepping “eliminates non-physical behavior.” The Step 6 report adds the $105\times$ number. But the ablation study (Review report) proved that adaptive stepping is the “primary stability mechanism” — a stronger and more precise claim. These two findings together make Conclusion 2 the most complete and well-evidenced conclusion in the paper. They belong in one place.

6.3 How It Fits with the Step 6 Report

The Step 6 report Section 6.2 (Addition 2) already has a version of this text. The addition here is to extend it by two sentences incorporating the ablation result, creating a single complete Conclusion 2 that cites both the before/after numbers and the no-NN ablation. When writing V3, use the version below instead of the Step 6 version.

6.4 Exact Text to Add to V3

Full replacement for Conclusion 2 in Section 5 of V3

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% Full replacement for Conclusion 2 in Section 5 of V3
% (supersedes the version in Step 6 report Section 6.2):
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Adaptive sub-stepping is the primary stability mechanism:

Distance-based adaptive sub-stepping eliminates non-physical body ejections and reduces worst-case trajectory error by $10^5\times$ (from 161.7~AU to 1.54~AU at $dt=0.01\text{~yr}$). Without this mechanism, SIMON produces body ejections at certain timesteps regardless of the NN correction being active---confirmed by the before-adaptive run in which the NN was present but ejection still occurred at 161.7~AU. An ablation retaining adaptive stepping but removing the NN correction yields bounded orbits with 8.5\% larger final RMS error (1.67 vs 1.54~AU), confirming that adaptive sub-stepping is the stability-critical component while the NN provides force accuracy. The mechanism activates on only 0.2--1.1\% of macro steps at a cost of 35\% additional per-step compute time (17.9 \to 24.3~ μs).

PART C — Addition to: Review Mentor Report

The Review report addresses five issues from an external reviewer. Its Section 7 (Complete V3 Paper Modification Checklist) covers all five issues. One writing task from the SIMON Optimization Report (Gap G3) was not incorporated: moving the NN invocation rate number into the Results section and providing exact text for this.

7. Addition C1: NN Invocation Rate Moved into Results (G3)

7.1 What Is Missing

The SIMON Optimization Report’s Gap G3 flagged that the 1.1% NN invocation rate number appears in V2’s Conclusion 6 but should also appear explicitly in the Results section (Section 4), where quantitative findings are presented. Currently, a reader of Section 4 cannot find out how often the NN is actually used — they have to search Conclusions for it.

The Review report addresses the framing of this number in Issues 1 and 2 (NN over-defensiveness, speed explanation), and provides the ablation result and speed breakdown. But it does not provide the specific sentence to add to Section 4.1 (Trajectory Overlay / Hybrid Usage) that states the invocation rate in context.

7.2 Why It Matters

The SIMON Optimization Report explicitly identified this as a structural problem: invocation rate data belongs in Results, not just Conclusions. A reviewer reading Section 4 expects to know how often the NN fires. Without it there, the Results section is incomplete. The fix is simple — one sentence — but it needs to be placed correctly and worded to match the ablation’s reframing.

7.3 How It Fits with the Review Report

The Review report’s Section 7 checklist includes “Issue 1: Ablation result — Add ablation table + one paragraph citing Run A, C, D results” as a new Section 4.6. The invocation rate sentence should be added to that same new section, immediately before the ablation table. Together they give the reader a complete picture: how often the NN fires (1.1%) and what happens when it doesn’t (8.5% larger error, still bounded).

7.4 Exact Text to Add to V3

Add as opening of new Section 4.6 in V3 (before ablation table)
% Add to the new Section 4.6 (Ablation Study) proposed in the Review report, % as the opening paragraph before the ablation table: The NN correction is invoked for 1.1\% of pair evaluations – exclusively at close encounters where pairwise separation falls below 0.15-AU. For the remaining 98.9\% of evaluations, exact unsoftened Newtonian gravity is used directly, bypassing the NN entirely. To quantify the

contribution of this targeted correction, we ran an ablation removing the NN while retaining all other mechanisms (adaptive sub-stepping, identical timestep, same integration loop). The results are summarised in Table~\ref{tab:ablation}.

% Then follow with the ablation table from Review report Section 4.3

% (Runs A, C, D with adaptive/NN on/off combinations).

8. Complete Integration Checklist

When writing V3, apply the following additions in this order. Each item references which mentor report it supplements.

#	Addition	V3 Location	Supplements	Size
A1	Strong central claim + generalizable principle	Section 5 (Conclusions, first bullet) Section 1 (last paragraph)	Revised Phase 1, Section 7.1	~60 words
A2	Section 3.1 research question update	Section 3.1 (replace existing question)	Revised Phase 1, Section 7.2	~80 words replaced
A3	Strong introduction sentence	Section 1 (final paragraph, new)	Revised Phase 1, Section 7.3	~80 words added
B1	CPU vs GPU explanation (G1)	Section 5, Conclusion 5 (replace)	Step 6, Section 6.2	~120 words replaced
B2	Conclusion 2 full expansion (105× + ablation)	Section 5, Conclusion 2 (replace)	Step 6, Section 6.2 Addition 2	~130 words replaced
C1	NN invocation rate into Results (G3)	New Section 4.6 (before ablation table)	Review report, Section 7 checklist	~80 words added

Note on Abstract (Step 9): The abstract is the final writing task and has been intentionally deferred to Step 9. It should be written last, after all V3 paper sections are finalised, so it accurately reflects the complete paper. The Revised Phase 1 report's Section 7.2 contains a template; the Step 6 report's Section 6.3 contains a recommended framing sentence. Both should be used as inputs when writing the abstract.

End of Additional Text Changes · SIMON V3 Research · April 28, 2026